

PACIFIC PRECISION GENOMICS LAB (SYNTHETIC FACILITY)

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Multi-Cancer Early Detection Screening Report

Next-Generation Sequencing cfDNA Methylation Profiling (Synthetic Demonstration)

Patient Information

Name: Robert T. Sample
DOB: 11/23/1968
Sex: Male
Patient ID: SYN-MRN-8849102

Provider Information

Provider: Dr. Marcus Vance
Facility: Bay Area Health System
NPI: 9982736450
Location: Fictional City, CA

Specimen Information

Specimen ID: SYN-SPEC-MCED-4491
Collection Date: Aug 1, 2026
Received Date: Aug 2, 2026
Report Date: Aug 6, 2026
Specimen Type: Blood / Plasma

Result: Cancer Signal Not Detected

Status: Final
Assay: cfDNA Methylation NGS

Test Result Summary: No cancer signal was detected in this synthetic cell-free DNA (cfDNA) blood sample. Methylation patterns analyzed across target genomic regions are consistent with non-malignant profiles. This result does not completely rule out the presence of all types of malignancy.

Laboratory Observations & Biomarkers

Test / Analyte	LOINC	Result	Reference Range	Interpretation
Cancer Signal Status	94076-7	Cancer Signal Not Detected	Cancer Signal Not Detected	Normal
Predicted Cancer Signal Origin 1	94077-5	None Detected	None	Normal
cfDNA Concentration	Local	8.4 ng/mL	5.0 - 25.0 ng/mL	Normal
Sequencing QC Pass Rate	Local	99.4%	>= 95.0%	Normal

Clinical Interpretation

A 'Cancer Signal Not Detected' result indicates that the targeted cell-free DNA methylation signatures associated with 50+ cancer types were not identified at actionable thresholds in this specimen. Routine preventative health examinations and age-appropriate standard screening guidelines (such as colonoscopy, mammography, and cervical cytology) should continue as medically indicated.

Methodology & Limitations

Methodology: Plasma cfDNA is extracted and treated with targeted bisulfite conversion followed by deep Next-Generation Sequencing (NGS) on Illumina NovaSeq systems. Epigenetic methylation patterns are evaluated using proprietary machine-learning classification models calibrated across multi-organ tissue datasets.

Limitations: A negative result does not completely eliminate the risk of cancer. Certain indolent tumors or cancers shedding low levels of cfDNA may not be detected. This test is validated for adult screening and is not intended for recurrence monitoring.